

Semester	B.E. Semester VII – Computer Engineering
Subject	Scalable ML & BDA Lab
Subject In-charge	Prof. SDW

Student Name	Diksha Parulekar.
Roll Number	23102A0068.
Grade and Subject Teacher's Signature	

Experiment Number	03	
Experiment Title	Multi-Source Retail Sales Data Integration and Analysis	
Resources / Apparatus Required	Hardware:	Software:
Description	This experiment focuses on integrating heterogeneous retail data from three different systems — sales transactions (CSV), product information (JSON), and customer information (Excel) — into a single consolidated dataset for analysis. The data was sourced from the UCI Online Retail Dataset, representing transaction-level sales records of a multinational retail company. The workflow began with importing each dataset using the appropriate R packages (readr for CSV, jsonlite for JSON, and readxl for Excel). Each dataset was inspected and cleaned individually — records with missing CustomerID, InvoiceNo, or StockCode were removed as they could not be reliably linked across sources; duplicate transaction records were dropped; and rows with invalid or zero Quantity and UnitPrice values were filtered out, since these represent returns, cancellations, or data entry errors rather than genuine sales. A new Revenue attribute was derived as $\text{Quantity} \times \text{UnitPrice}$ to enable financial analysis. The cleaned datasets were then integrated using dplyr join operations. An inner_join() was used to merge transactions with product data, since a transaction without valid product/price information carries no analytical value. A left_join() was then used to merge in customer data, preserving all valid sales transactions even where a customer record was incomplete, so that revenue figures were not understated. Using group_by() and summarise(), the integrated dataset was analyzed to determine total revenue, the top 5 products, top 5 countries, and top 5 customers	

by revenue contribution. Customers were further segmented into Low, Medium, High, and Premium value tiers using case_when() based on their total purchase value. Finally, the cleaned and integrated dataset was exported into a SQLite database (retail_sales.db) as the retail_sales table, and two SQL queries were executed directly from R to validate that the data was correctly stored and queryable.

Program & Output

```
install.packages(c("tidyr", "readr", "janitor", "readxl", "rvest", "httr", "sqldf", "lubridate"), quiet = TRUE)

library(tidyr)
library(readr)
library(janitor)
library(readxl)
library(rvest)
library(httr)
library(sqldf)
library(lubridate)

cat("Packages loaded successfully!\n")

# Attaching package: 'tidyr'
# The following objects are masked from 'package:stats':
#   filter, lag
# The following objects are masked from 'package:base':
#   intersect, setdiff, setequal, union

# Attaching package: 'lubridate'
# The following objects are masked from 'package:base':
#   date, intersect, setdiff, union

Packages loaded successfully

set.seed(42)

# Try downloading the UCI Online Retail dataset
url_url <- "https://archive.ics.uci.edu/ml/machine-learning-databases/online-retail/uci.zip"
temp_file <- "UCIOnlineRetail.zip"

download_ok <- tryCatch({
  download.file(url_url, temp_file, mode = "wb", quiet = TRUE)
}, error = function(e) FALSE)

if (download_ok && file.exists(temp_file) && file.size(temp_file) > 1000) {
  raw_data <- read_excel(temp_file)
  cat("UCI dataset downloaded successfully. Rows:", nrow(raw_data), "\n")
} else {
  cat("UCI dataset failed - generating synthetic online retail-style data!\n")
  # ... (code for generating synthetic data) ...
}

data <- intersect, setdiff, union

Packages loaded successfully

set.seed(42)

# Try downloading the UCI Online Retail dataset
url_url <- "https://archive.ics.uci.edu/ml/machine-learning-databases/online-retail/uci.zip"
temp_file <- "UCIOnlineRetail.zip"

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  raw_data <- read_excel(temp_file)
  cat("UCI dataset downloaded successfully. Rows:", nrow(raw_data), "\n")
} else {
  cat("UCI dataset failed - generating synthetic online retail-style data!\n")
  # ... (code for generating synthetic data) ...
}

# transactions.csv
transactions_raw <- raw_data %>%
  select(invoiceid, stockcode, customerid, quantity, invoicedate) %>%
```

UCI dataset downloaded successfully. Rows: 342989

InvoiceId	stockcode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
<chr>	<chr>	<chr>	<dbl>	<dttm>	<dbl>	<dbl>	<chr>
SN306	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	17850	United Kingdom
SN306	71803	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.56	17850	United Kingdom
SN306	84406B	CREAM CUPRO HEARTS COAT HANGER	6	2010-12-01 08:26:00	2.75	17850	United Kingdom
SN306	84030	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 08:26:00	3.39	17850	United Kingdom
SN306	84030E	RED WOOLLY HOTTIE WHITE HEART	6	2010-12-01 08:26:00	3.39	17850	United Kingdom
SN306	22782	SET 7 BABUSHKA NESTING BOXES	2	2010-12-01 08:26:00	7.65	17850	United Kingdom

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File Edit View Insert Runtime Tools Help

Commands Code Text Run all

Files

- sample_data
- individual_raw_data
- customers.xlsx
- final_integrated_retail_data.csv
- products.json
- retail_sales.db
- transactions.csv

transactions.csv

```

# transactions.csv
transactions_raw <- read_data %>%
  select(invdate, stockcode, customerid, quantity, invoicedate) %>%
  distinct()
write_csv(transactions_raw, "transactions.csv")

# products.json
products_raw <- read_data %>%
  select(stockcode, description, unitprice) %>%
  distinct(stockcode, keep_all = TRUE)
write_json(products_raw, "products.json")

# customers.xlsx
customers_raw <- read_data %>%
  filter(!is.na(customerid)) %>%
  select(customerid, country) %>%
  distinct(customerid, keep_all = TRUE)
write_excel(customers_raw, "customers.xlsx")

cat("Source files created!\n")
cat("transactions.csv", nrow(transactions_raw), "rows\n")
cat("products.json", nrow(products_raw), "rows\n")
cat("customers.xlsx", nrow(customers_raw), "rows\n")

# Source files created:
# transactions.csv: 350498 rows
# products.json: 4079 rows
# customers.xlsx: 4372 rows

```

Task 1 - Import and Clean the Data

```

# Import from the three sources
transactions <- read_csv("transactions.csv", show_col_types = FALSE)
products <- read_json("products.json") %>% select(stockcode, description, unitprice)
customers <- read_excel("customers.xlsx")

cat("Raw counts -> transactions:", nrow(transactions),
    "\n products:", nrow(products),
    "\n customers:", nrow(customers), "\n")

# --- Cleaning Transactions ---
transactions_clean <- transactions %>%
  filter(!is.na(customerid), !is.na(invdate), !is.na(stockcode)) %>%
  filter(quantity > 0)

# remove missing keys
# remove invalid quantity

# --- Cleaning Products ---
products_clean <- products %>%
  filter(!is.na(stockcode), !is.na(description), !is.na(unitprice)) %>%
  filter(unitprice > 0)

# remove invalid unit price

# --- Cleaning Customers ---
customers_clean <- customers %>%
  filter(!is.na(customerid), !is.na(country)) %>%
  distinct(customerid, keep_all = TRUE)

cat("Cleaned counts -> transactions:", nrow(transactions_clean),
    "\n products:", nrow(products_clean),
    "\n customers:", nrow(customers_clean), "\n")

cat("\nCleaning decisions:\n")
cat("1. Removed rows with missing CustomerID/Invoice/Stockcode (can't be linked)\n")
cat("2. Removed exact duplicate transaction rows\n")
cat("3. Removed transactions with quantity == 0 (returns/invalid entries)\n")
cat("4. Removed products with unitprice == 0 or missing (data entry errors)\n")

Raw counts -> Transactions: 350498 | Products: 4079 | Customers: 4372
Cleaned counts -> Transactions: 302398 | Products: 3885 | Customers: 4372

Cleaning decisions:
- Removed rows with missing CustomerID/Invoice/Stockcode (can't be linked)
- Removed exact duplicate transaction rows
- Removed transactions with quantity == 0 (returns/invalid entries)
- Removed products with unitprice == 0 or missing (data entry errors)

```

Task 2 - Integrate the Multiple Data Sources

```

# Join transactions -> products (inner join: only keep transactions with valid product info)
integrated <- transactions_clean %>%
  inner_join(products_clean, by = "stockcode")

# Join with customers (left_join: keep all transactions even if some customer info missing)
integrated <- integrated %>%
  left_join(customers_clean, by = "customerid")

# Add Revenue column
integrated <- integrated %>%
  mutate(revenue = quantity * unitprice)

cat("Final integrated dataset dimensions:", dim(integrated)[1], "rows x", dim(integrated)[2], "columns\n")

# Unmatched records check
unmatched_records <- integrated %>% filter(!is.na(customerid)) %>%
  select(customerid, country) %>%
  distinct(customerid, keep_all = TRUE)

cat("Unmatched records:\n")
cat("1. inner_join used for products: a transaction is meaningless without valid product/price data\n")
cat("2. left_join used for customers: we retain all valid sales transaction even if a customer record is missing, to avoid losing revenue data\n")

head(integrated)

# final integrated dataset dimensions: 302398 rows x 9 cols
# Records with unmatched customer/country info: 0

Join Justification:
- inner_join used for products: a transaction is meaningless without valid product/price data
- left_join used for customers: we retain all valid sales transaction even if a customer record is missing, to avoid losing revenue data

```

stockcode	customerid	quantity	invdate	description	unitprice	country	revenue
530365	854334	17550	6 2010-12-01 08:26:00	WHITE HANGING HEART T-LIGHT HOLDER	2.55	United Kingdom	15.30
530365	71953	17550	6 2010-12-01 08:26:00	WHITE METAL LANTERN	3.39	United Kingdom	26.34
530365	844068	17550	8 2010-12-01 08:26:00	CREAM CUPID HEARTS COAT HANGER	2.75	United Kingdom	22.80
530365	840290	17550	6 2010-12-01 08:26:00	KNITTED UNION FLAG HOT WATER BOTTLE	3.39	United Kingdom	26.34
530365	840296	17550	6 2010-12-01 08:26:00	RED WOOLLY HOTTIE WHITE HEART	3.39	United Kingdom	26.34
530365	22752	17550	2 2010-12-01 08:26:00	SET 7 BABUSHKA NESTING BOXES	7.85	United Kingdom	15.30

Task 3 - Sales and Customer Analysis

```

# 1. Total sales revenue
total_revenue <- sum(integrated$revenue, na.rm = TRUE)
cat("Total Sales Revenue:", format(total_revenue, 0), "\n")

# 2. Top 5 products by revenue
top_products <- integrated %>%
  group_by(stockcode, description) %>%
  summarize(total_revenue = sum(quantity * unitprice, na.rm = TRUE)) %>%
  arrange(desc(total_revenue))

```

The screenshot displays a Jupyter Notebook interface with two pages. The first page contains Python code for data analysis and summary statistics, while the second page contains SQL queries and their results.

Page 1: Data Analysis and Summary Statistics

```

errange(TotalRevenue) %>%
  head(1)

cat("High-performing market:", best_market,
    "\n- Highest total revenue, indicating strong customer base and demand")
cat("Underperforming market:", worst_market,
    "\n- Lowest revenue, indicating limited reach or low purchase frequency")

# Summary Statistics
Total Sales Revenue: 38752948

Top 5 Products by Revenue:
# A tibble: 5 x 3
  stockCode description      TotalRevenue
  <int>   <chr>          <dbl>
1  22843  AUREL CHIFF_ LITTLE REDDIE  224796.
2  47866  PATTY BATHING  125796.
3  22823  REMOXY CLOSETING 3 TIER  123696.
4  225238  WOLFE WASHING MESH T-LIGHT HOLDER  22796.
5  22566  HODAR CLOSETOP FOLDING BAR  22696.

Top 5 Countries by Revenue:
# A tibble: 5 x 2
  Country      TotalRevenue
  <chr>          <dbl>
1 United Kingdom  883387.1
2 Netherlands  333884.
3 EUNE  223486.
4 Germany  223379.
5 France  222879.

Top 5 Customers by Revenue:
# A tibble: 5 x 2
  CustomerID TotalRevenue
  <int>          <dbl>
1  18140  228708.
2  18464  172751.
3  12408  228898.
4  18713  228771.
5  18446  228472.

Customer Segment distribution:
High Value  Low Value Medium Value  Premium
      932         117         1209         1881

High-performing market: United Kingdom - Highest total revenue, indicating strong customer base and demand
Underperforming market: Saudi Arabia - Lowest revenue (181.4k), suggesting limited reach or low purchase frequency
  
```

Task 4 - Store and Retrieve Data Using SQL

```

# Create SQLite database
con <- dbConnect(SQLite::SQLite(), "retail_sales.db")

# Import cleaned/processed dataset
dbWriteTable(con, "retail_sales", integrated, overwrite = TRUE)
cat("Table 'retail_sales' created with", dbGetQuery(con, "SELECT COUNT(*) as n FROM retail_sales"), "rows\n")

# Query 1: Top 5 customers by revenue
  
```

Page 2: SQL Queries and Results

```

# Query 1: Top 5 customers by revenue
con <- dbConnect(SQLite::SQLite(), "retail_sales.db")
dbGetQuery(con, "SELECT CustomerID, TotalRevenue FROM retail_sales ORDER BY TotalRevenue DESC LIMIT 5")

# Query 2: Total Revenue by Country
dbGetQuery(con, "SELECT Country, SUM(TotalRevenue) as TotalRevenue FROM retail_sales GROUP BY Country")

# Query 3: Top 5 Products by Revenue
dbGetQuery(con, "SELECT stockCode, description, TotalRevenue FROM retail_sales ORDER BY TotalRevenue DESC LIMIT 5")

# Query 4: Customer Segment distribution
dbGetQuery(con, "SELECT High Value, Low Value, Medium Value, Premium FROM retail_sales")
  
```

The second page shows the results of these SQL queries, including a list of countries and their total revenue, and a list of products and their total revenue.

	Database connection closed. Retail_sales.db saved. <pre> [8] ✓ Os cat("=== BUSINESS INSIGHTS ===\n\n") cat("1. Revenue is highly concentrated in a few top products and countries,\n") cat(" suggesting the business should prioritize inventory and marketing spend\n") cat(" around these high-performing categories and regions.\n\n") cat("2. A significant share of customers fall into the 'Low Value' segment,\n") cat(" indicating an opportunity for targeted upselling or loyalty programs\n") cat(" to move them toward 'Medium' and 'High Value' tiers.\n\n") cat("3. The gap between the top-performing and underperforming markets highlights\n") cat(" an expansion opportunity – underperforming regions may benefit from\n") cat(" localized promotions or better product-market fit analysis.\n") # Save cleaned integrated dataset for submission write_csv(integrated, "final_integrated_retail_data.csv") cat("\nFinal dataset exported as final_integrated_retail_data.csv\n") cat("Files ready for submission: transactions.csv, products.json, customers.xlsx,\n") cat("final_integrated_retail_data.csv, retail_sales.db\n") </pre> ▼ <pre> *** === BUSINESS INSIGHTS === 1. Revenue is highly concentrated in a few top products and countries, suggesting the business should prioritize inventory and marketing spend around these high-performing categories and regions. 2. A significant share of customers fall into the 'Low Value' segment, indicating an opportunity for targeted upselling or loyalty programs to move them toward 'Medium' and 'High Value' tiers. 3. The gap between the top-performing and underperforming markets highlights an expansion opportunity – underperforming regions may benefit from localized promotions or better product-market fit analysis. Final dataset exported as final_integrated_retail_data.csv Files ready for submission: transactions.csv, products.json, customers.xlsx, final_integrated_retail_data.csv, retail_sales.db </pre>
Conclusion :	This experiment demonstrated the complete pipeline of multi-source data integration and analysis using R — from importing heterogeneous file formats (CSV, JSON, Excel), through cleaning and standardizing inconsistent data, to joining datasets using dplyr and storing the final result in a relational database (SQLite) for downstream querying. The use of inner_join() and left_join() illustrated how join strategy should be chosen based on business context — ensuring no valid revenue data is lost while still enforcing referential integrity where required (e.g., valid product information). The case_when()-based customer segmentation and country-level revenue analysis provided actionable business insight into customer value distribution and market performance. Overall, the experiment reinforced practical skills in data cleaning, multi-source integration, dplyr-based analytics, and SQL connectivity within R — all essential competencies for real-world data analyst workflows.
Business Insights	- Revenue is concentrated among a small number of top-performing products and countries. This indicates that the business should prioritize inventory planning, stock availability, and marketing spend around these high-revenue categories and regions to maximize returns. - A significant proportion of customers fall into the "Low Value" segment. This presents an opportunity for targeted upselling, personalized offers, and loyalty programs to gradually shift these customers into the "Medium" and "High Value" tiers, increasing overall customer lifetime value. - There is a notable revenue gap between the top-performing and underperforming markets. This suggests an expansion opportunity in the

	weaker markets — through localized promotions, better product-market fit analysis, or region-specific pricing strategies — to close the performance gap.
--	--